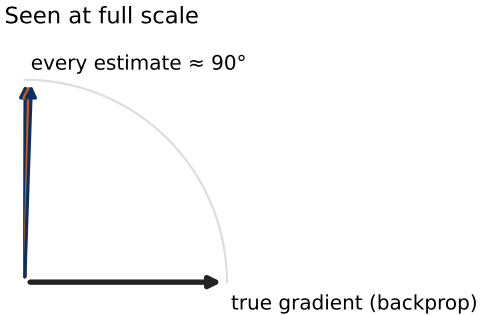
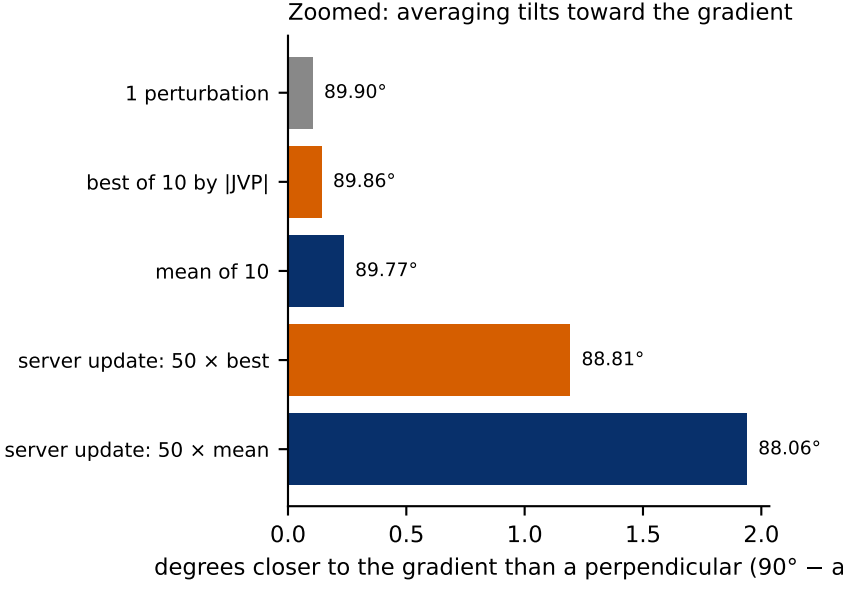

Seen at full scale
every estimate ≈ 90°
true gradient (backprop)


Zoomed: averaging tilts toward the gradient
1 perturbation
89.90°
best of 10 by |JVP|
89.86°
mean of 10
89.77°
server update: 50 × best
88.81°
server update: 50 × mean
88.06°
0.0
0.5
1.0
1.5
2.0
degrees closer to the gradient than a perpendicular (90° − a